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Are You Getting What You Pay For?

Auditing Model Substitution in LLM APIs:

Why Software Tests Fail and Hardware Attestation Wins

Commercial LLM providers have strong economic incentives to covertly substitute cheaper models — and today's detection methods cannot stop them.

Economic Incentives Drive Covert Swaps

Providers charge for named models (e.g. Llama-3-70B, Qwen2-72B) but the black-box API gives users no guarantee the advertised model is actually served. Covert substitution reduces GPU costs while maintaining pricing.

Hypothesis Testing Framework: Can an auditor distinguish P_spec(y|x) from P_actual(y|x)? (H_0: same model; H_1: substituted model)

↓

Smaller Model

e.g., 8B instead of 70B

≈

Quantized Variant

FP8 / INT8 instead of FP16

✦

Fine-Tuned Version

Updated or modified weights

≠

Different Family

Entirely different architecture

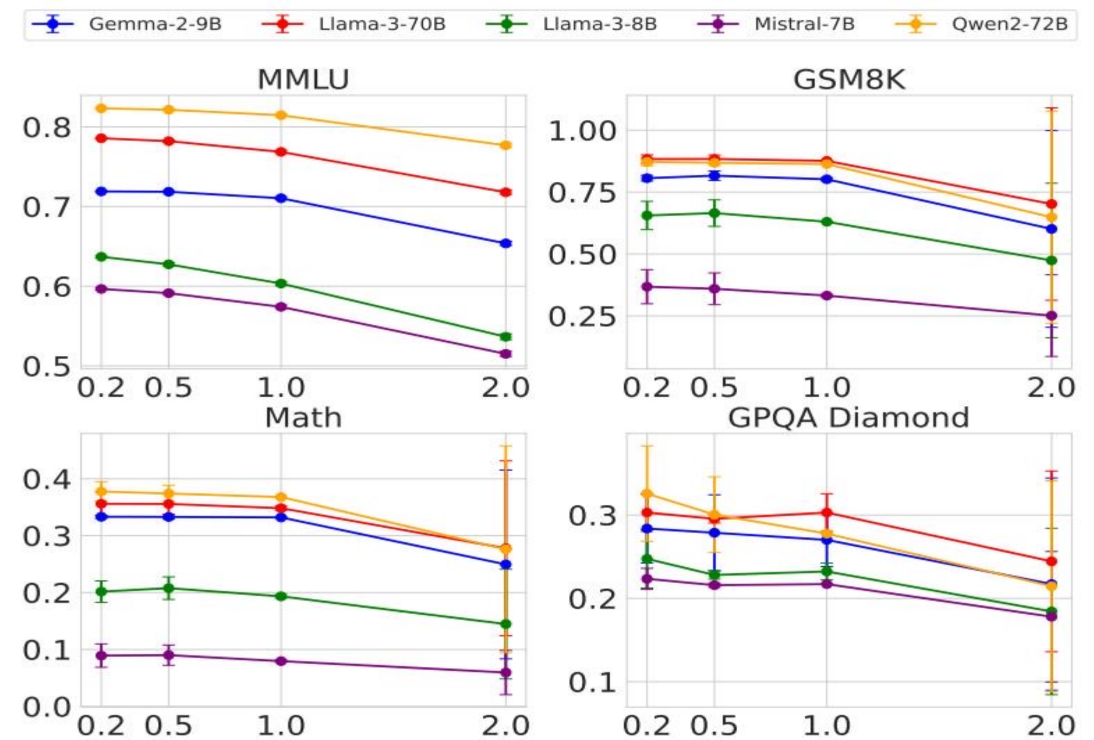
FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to Detect

Performance drops are small, often within error bars — FP8 is an attractive near-undetectable cost-cutting measure.

Model	MMLU		GSM8K		MATH		GPQA Diamond	
	FP16	FP8	FP16	FP8	FP16	FP8	FP16	FP8
Llama-3-8B-Instruct	62.69	62.43	61.14	60.90	20.65	**14.91**	22.62	**20.14**
Llama-3-70B-Instruct	78.05	77.88	88.06	87.35	35.69	35.75	29.60	33.12
Gemma-2-9b-it	71.86	71.92	81.80	79.41	33.41	32.53	28.64	27.81
Qwen2-72B-Instruct	82.18	81.98	86.72	86.82	37.39	37.67	29.93	31.08
Mistral-7B-Instruct-v0.3	59.15	58.77	35.90	**32.20**	8.94	7.68	21.60	22.72

Green = within/near error bars **Red/Bold** = notable drop *FP8 scores often indistinguishable from FP16 → detection is near-impossible*

Higher Temperature Amplifies Performance Gaps but Increases Noise



Temperature 0.2 – 1.0

Models maintain expected performance; FP8 and FP16 scores remain close, signal is weak.

Temperature 2.0

Performance degrades across all models. Gaps become noisy and unreliable for detection.

Key Implication

Higher temperature amplifies noise faster than signal, making statistical tests even less reliable for auditing.

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

The Idea:

Compare token-level log probabilities between a local reference model and the API response to detect substitutions.

Why It Fails: Production Nondeterminism

1. SW framework: transformers 4.40 vs 4.50 → different log-probs
2. Inference engine: vLLM 0.6.2 vs 0.8.2 → different log-probs
3. GPU hardware: A100 vs H100 → different log-probs

Result: An auditor cannot distinguish a legitimate model served on different infrastructure from an actual substitution.

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Fig. 5: Log-prob variation across frameworks & hardware for Llama-3-8B

Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive

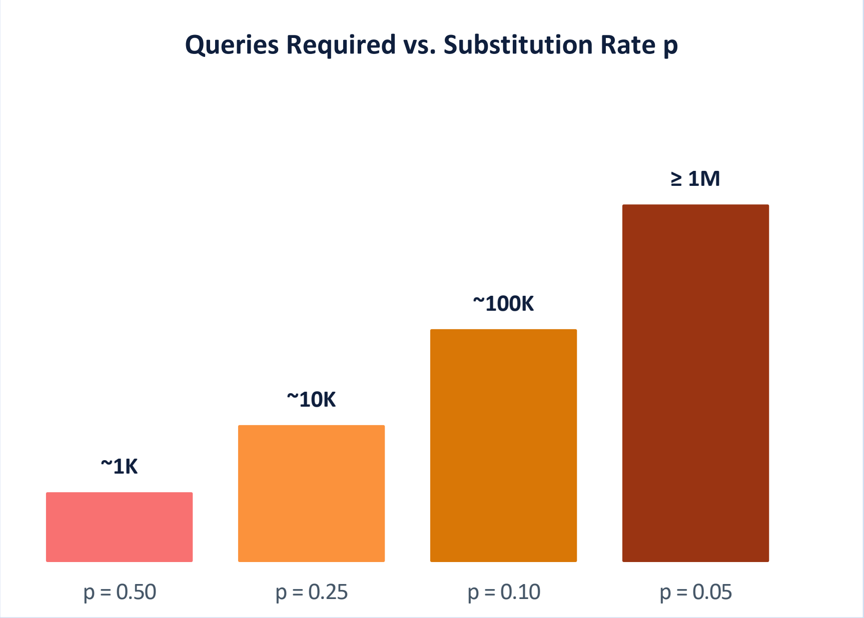
Providers mix substitute model responses with probability p. As p → 0, the mixed distribution converges to the specified model — requiring exponentially more queries to detect.

```
P_mixed(y|x) = (1−p)·P_spec(y|x) + p·P_sub(y|x)
```

Benchmark Evasion Attack

Providers detect known benchmark prompts
→ Route those to the genuine model
→ Serve substitutes for all other traffic

Result: Audits pass while users are deceived.



Query budget grows exponentially →

Software-Only Auditing Methods Are Fundamentally Unreliable

Detection Method	How It Works	Core Weakness	Practical Limitation
Text-Based Statistical Tests (e.g., MMD)	Sample API outputs; apply two-sample statistical tests to text distributions	FP8 scores overlap original within error bars; distributions too similar	Requires 100K–1M+ queries; fails against subtle or low-p substitutions
Log-Probability Fingerprinting	Compare token-level log-probs between local reference and API responses	Production nondeterminism: SW versions and GPU hardware alter log-probs	Cannot distinguish legitimate infra variation from actual model substitution
Adversarial Countermeasures (Provider)	Randomized routing (p fraction to substitute); detect & route benchmark prompts	Mixed distribution → P_spec as p → 0; benchmark evasion defeats specific tests	All black-box software auditing defeated by design — structural, not incidental

Verdict: Software-only detection is fundamentally query-prohibitive or defeatable — a structural, not an implementation, problem.

THE SOLUTION: HARDWARE ATTESTATION

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

TEEs are hardware-secured enclaves providing cryptographic attestation of the code and data being executed. Unlike ZKPs (computationally prohibitive for large models), TEEs are practical and deployable today via NVIDIA Confidential Compute on H100 GPUs.

1

Weights Loaded

Correct model weights loaded into hardware-secured enclave

2

Inference Runs

Computation executes inside TEE; provider cannot tamper

3

Proof Generated

Attestation report signed by hardware root of trust

4

Client Verifies

Auditor verifies proof without trusting the provider

Property	Software Tests	Zero-Knowledge Proofs	TEEs
Provability	None — defeatable	Provable in theory	Provable in practice
Compute Cost	Low but query-heavy	Prohibitive for LLMs	Modest overhead
Deploy Today	Unreliable	Not viable at scale	✓ NVIDIA H100 / CC
Adversarial Robustness	Defeated by design	Depends on prover	Hardware root of trust

KEY TAKEAWAYS

Close the Verification Gap with Hardware Attestation

01

Economically Motivated & Undetectable

Covert substitution reduces GPU costs while maintaining pricing. Black-box APIs give users no guarantee the advertised model is served.

02

FP8 Quantization Barely Affects Benchmarks

Scores fall within error bars on MMLU and GSM8K. Temperature scaling amplifies noise faster than detection signal.

03

Nondeterminism Defeats Log-Prob Fingerprinting

SW framework and GPU hardware (transformers, vLLM, A100 vs H100) produce different log-probs for the same model.

04

Randomized Routing Defeats Statistical Tests

Mixed distribution with $p \to 0$ requires exponentially more queries. Benchmark evasion makes audits trivially passable.

05

TEEs Are the Only Viable Path Today

NVIDIA Confidential Compute (H100) provides provable model integrity with modest overhead — actionable now.

Community Call to Action: Demand TEE-backed attestation from LLM API providers. Treat unverified inference as untrusted.

arxiv.org/abs/2504.04715